Computer Graphics Final Project 2026 Report

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Theme: AI Future World – Palmanova Star City

i. Project Setup

This section explains how to compile and run the project.

- **Prerequisites:** You must have a C++ Compiler (g++), GNU Make, and FreeGLUT installed on your system.
- **Configuration:** The project expects FreeGLUT to be located at `C:\freeglut` by default. If your installation is elsewhere, you must edit lines 3–4 of the Makefile to update the include and library paths.
- **Compilation:** Open your terminal and run `make` to compile the project.
- **Execution:** Run `make run` to launch the application.
- **Cleanup:** Run `make clean` to clear build files.

ii. User Guide

The application supports both Keyboard controls and Windows XInput Gamepad controls for navigating the 3D city.

Keyboard Navigation

Key	Action
W / S	Move Forward / Backward.
A / D	Strafe Left / Right.
Q / 0	Tilt Camera Up / Tilt Camera Down.
Left Arrow / Right Arrow	Pan Camera Left / Right.

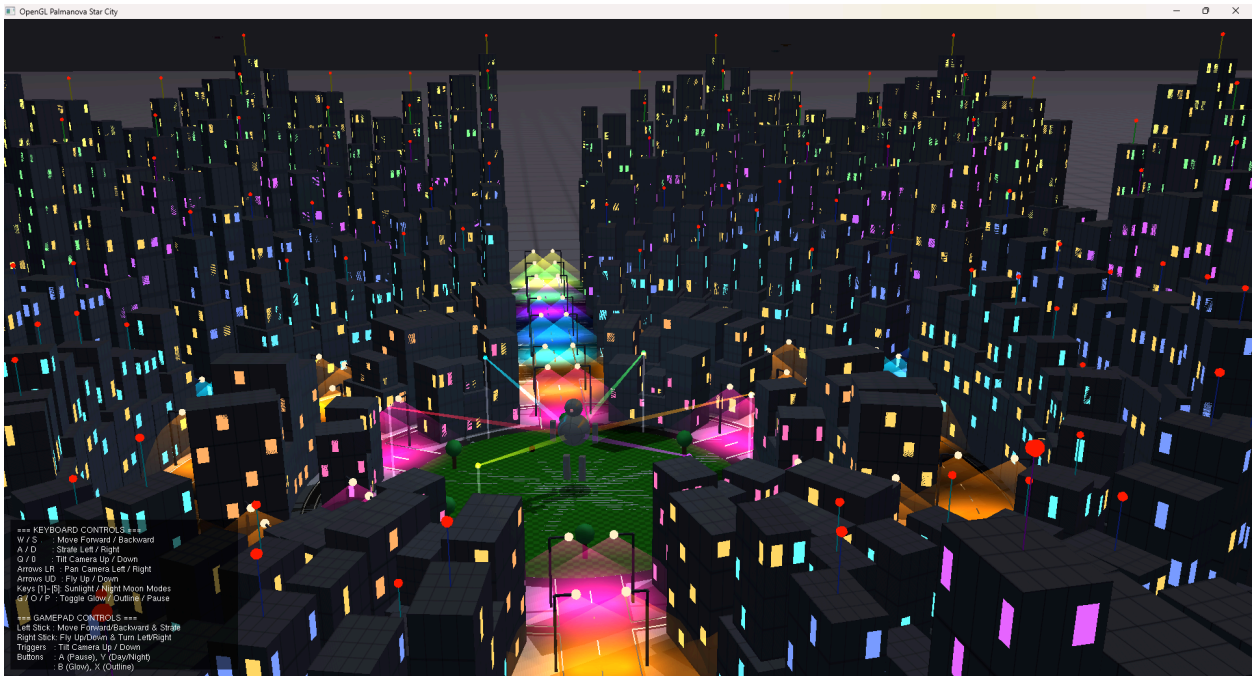
Key	Action
Up Arrow / Down Arrow	Fly Vertically Up / Down.
1 - 4	Activate Night Mode & switch between 4 different Moon positions.
5	Activate Daylight Mode (Bright Sunlight).
G	Toggle Skyscraper Windows & Spire Glow (Night only).
O	Toggle Skyscraper Neon Grid Outlines (Night only).
P	Toggle Pause / Play animations.

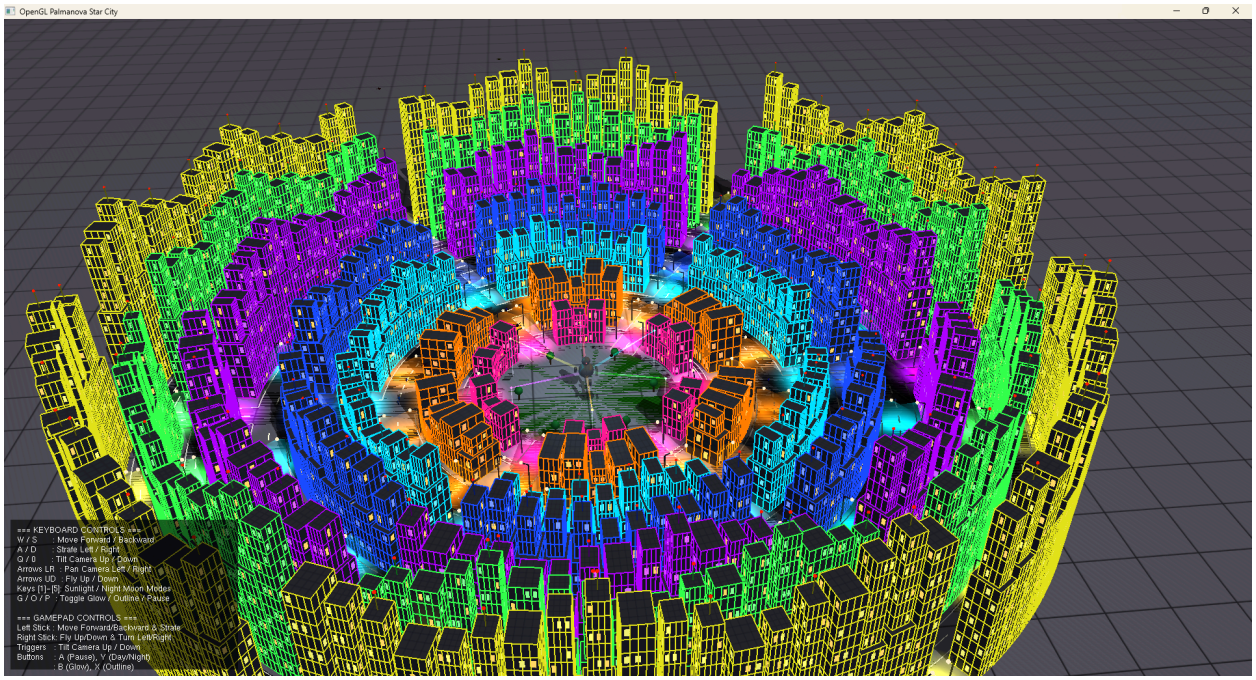
Gamepad Navigation (Xbox Controller)

Control	Action
Left Thumbstick Y / X	Move Forward / Backward and Strafe Left / Right.
Right Thumbstick Y / X	Fly Vertically Up / Down and Turn Left / Right.
Left & Right Triggers	Tilt Camera Up / Down.
Button A	Toggle Animation Pause.
Button Y	Toggle Day / Night Mode.
Button X	Toggle Skyscraper Glow.

Control	Action
Button B	Toggle Skyscraper Outlines.

iii. Screenshots





iv. Requirement Checklist

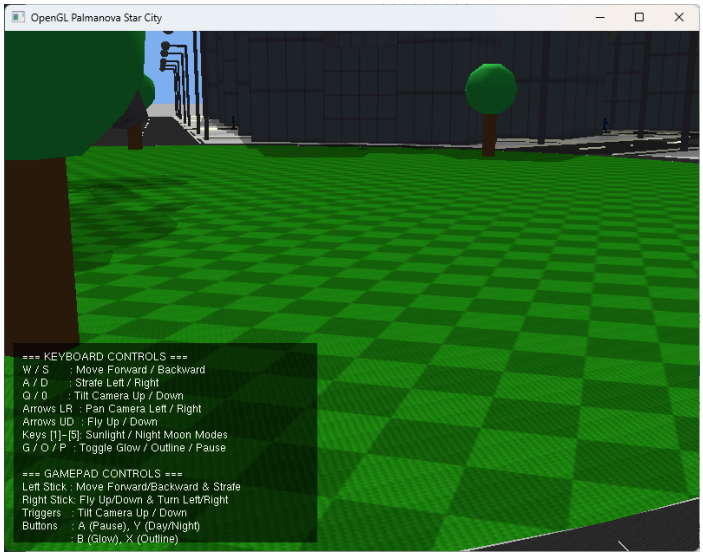
Below is the clear identification of which objects fulfill the project requirements. *(Note: Please insert screenshot evidence beneath each item before submitting)*.

- **1. Texture Mapping (10%)**:

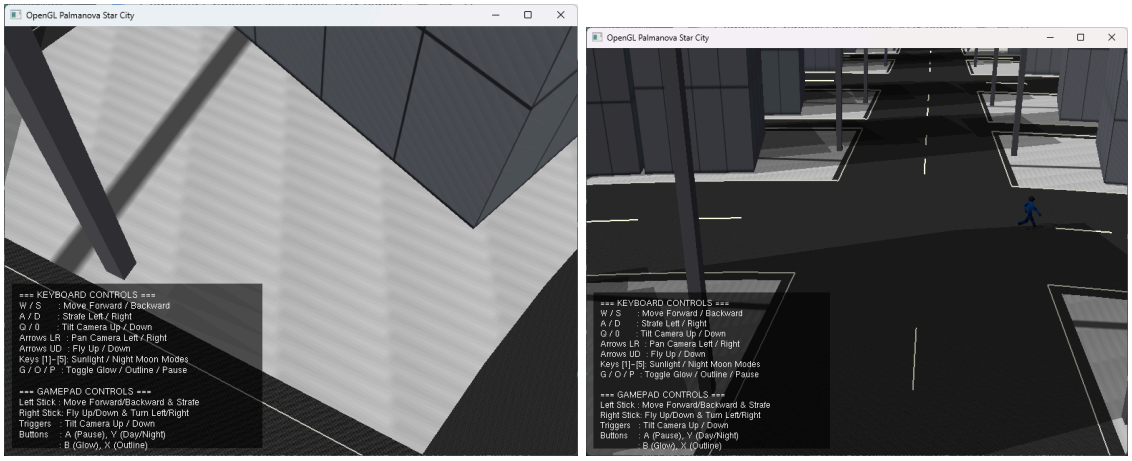
  The project utilizes 7 custom-generated textures, applying them to various object types.

  The textures include cyber grass, concrete, asphalt, metal, neon grids, holograms, and spotlights.

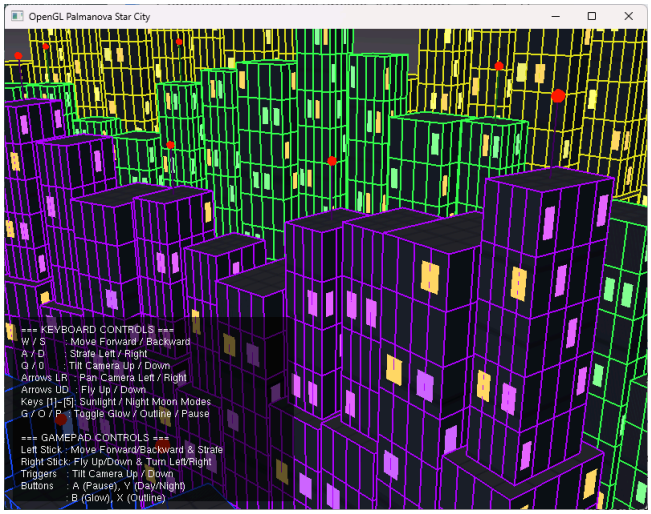
  The AI-related objects (pedestrians and drones) fulfill the specific AI texture requirement by using the hologram.tga and cyber_metal.tga textures.



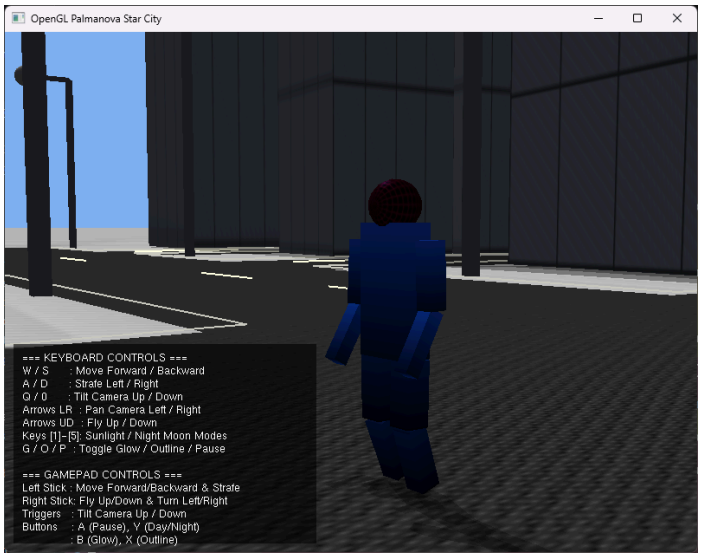
Texture: grass



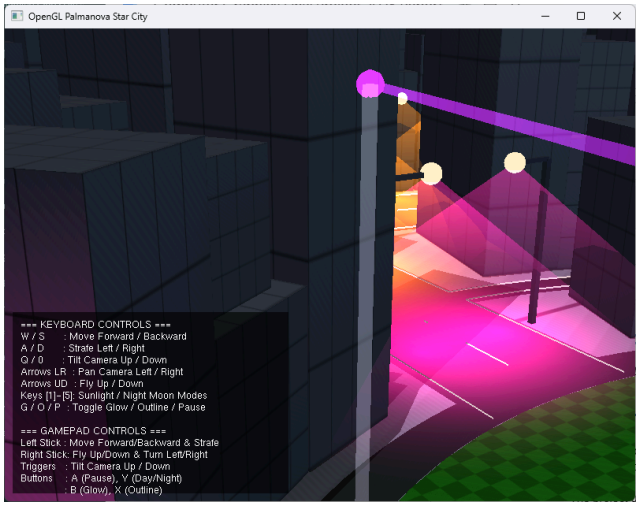
Texture: concrete, metal and asphalt



Texture: neon grid



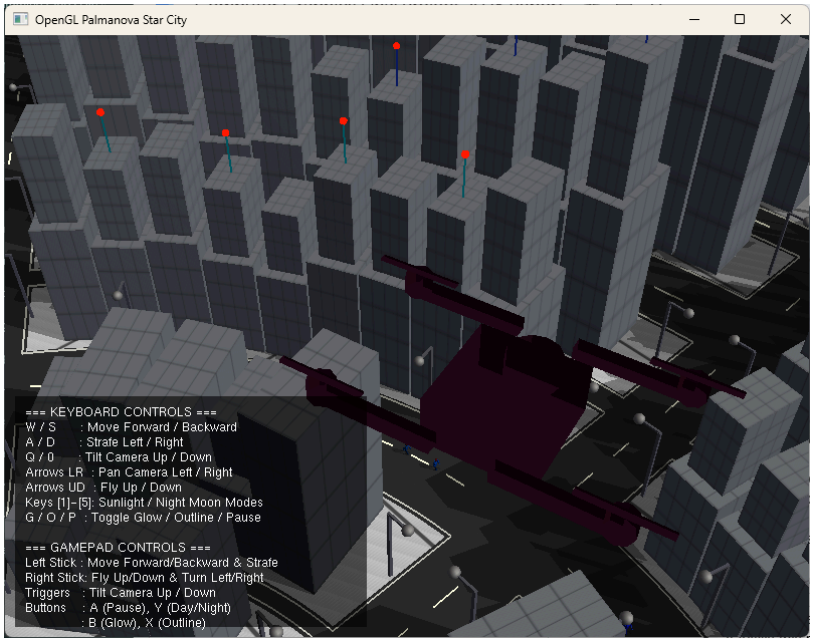
Texture: hologram



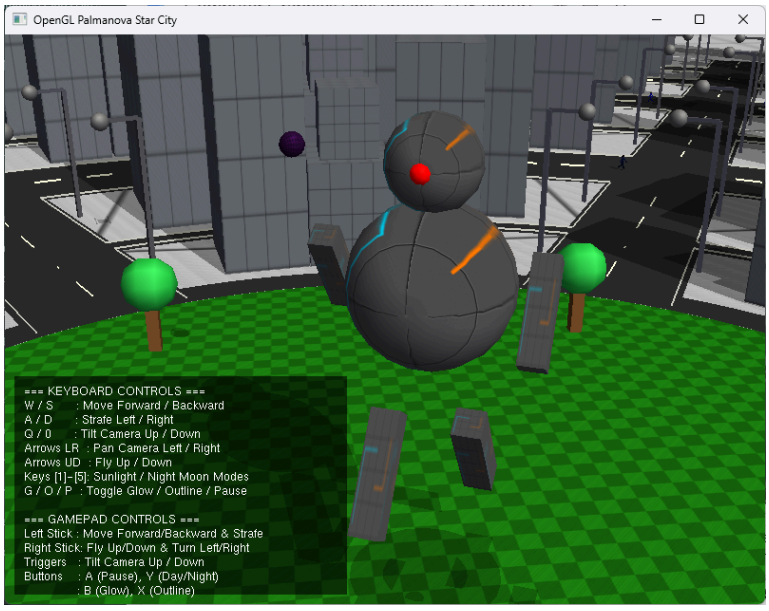
Texture: Spotlight

2. AI theme-related Object Model (10%):

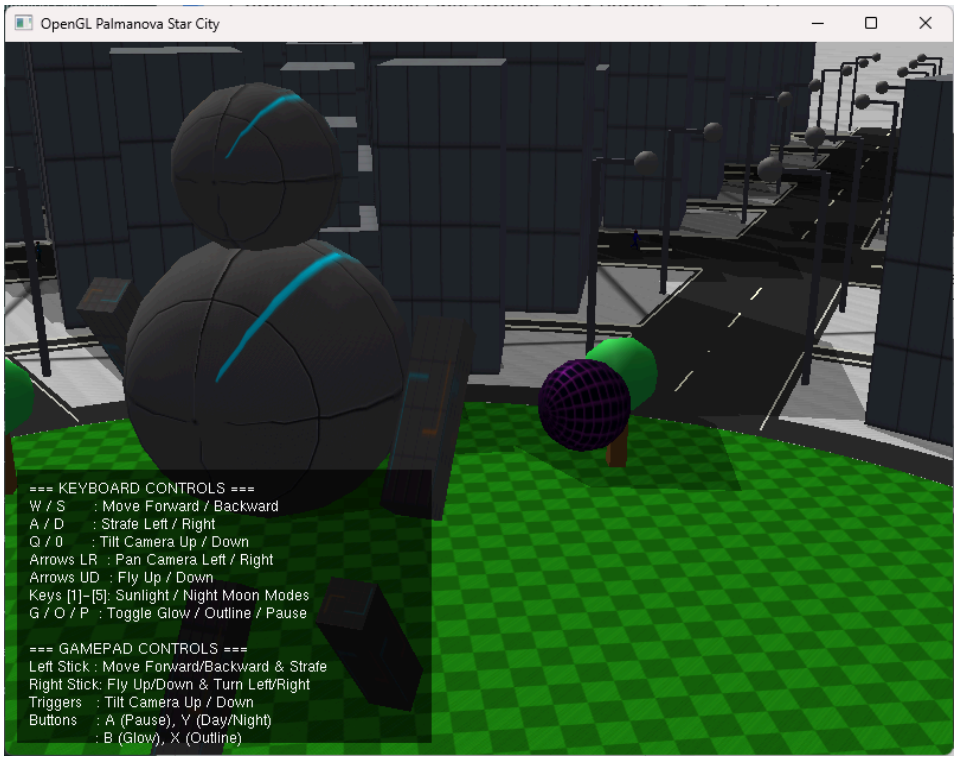
- The project loads a custom external drone.obj model, representing an AI delivery drone.
- There are 8 active drones rendered in the scene, which include altitude oscillation and body spin animations.



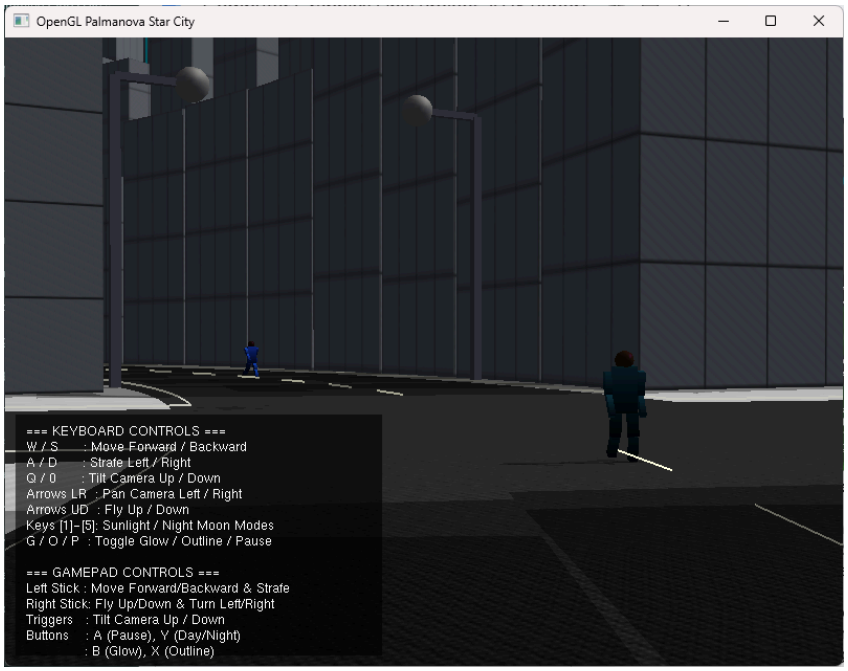
- **3a. Self Rotation (5%)**:
  - Object_A is a central giant humanoid robot located at the park center.
  - It fulfills this requirement by rotating continuously on its Y-axis.



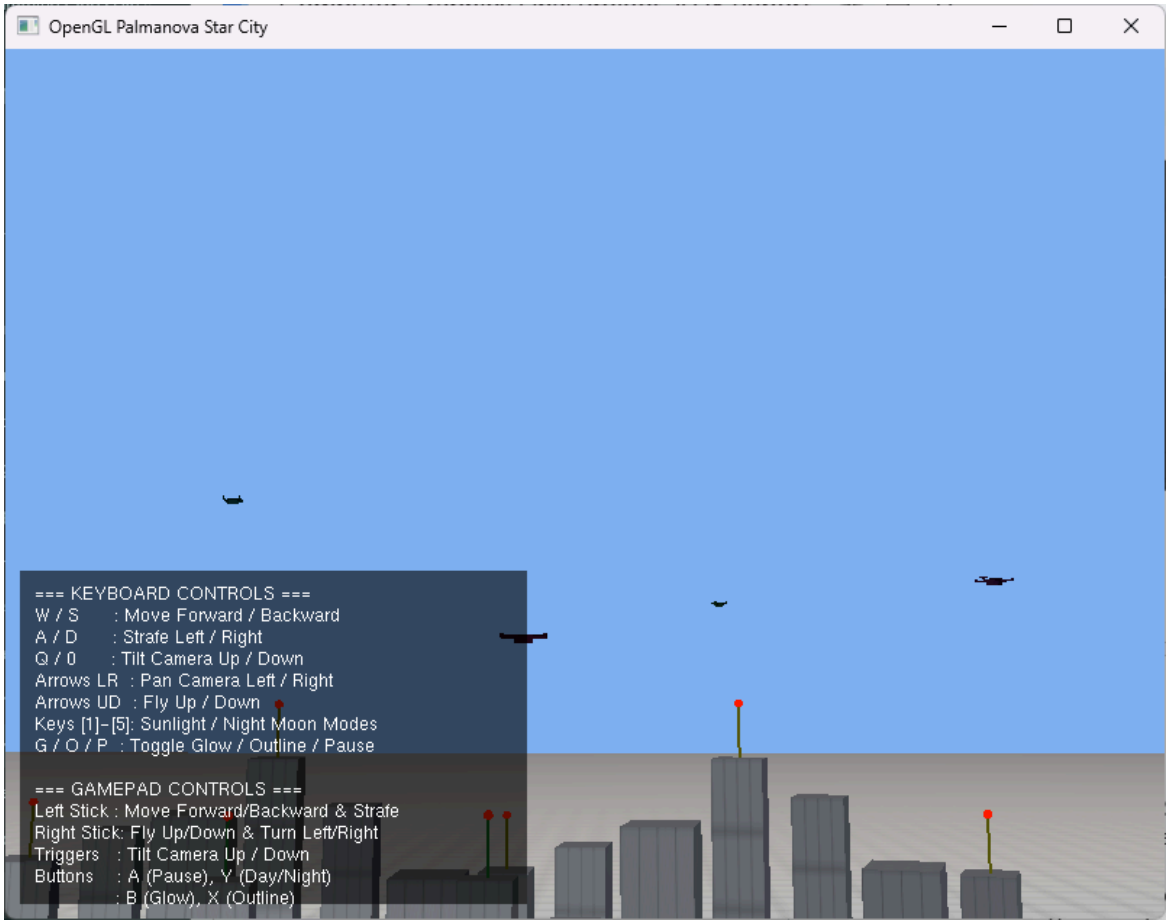
- **3b. Orbit Motion (5%)**:
    - Object_B is an orange surveillance satellite sphere.
    - It revolves around the central robot (Object_A) while oscillating vertically using a sine wave function.



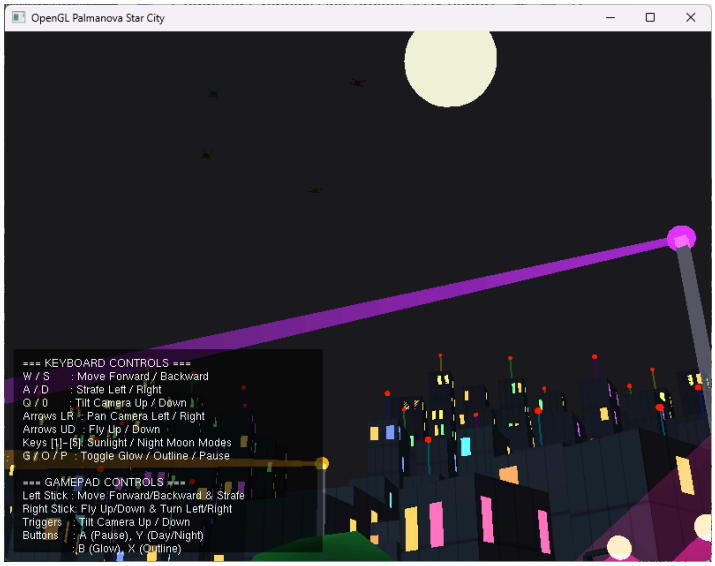
- **3c. Custom Animation (10%)**:
    - **Animation 1:** 30 robot pedestrians utilize hierarchical walking, swinging legs and arms in counter-phases using sine/cosine functions.



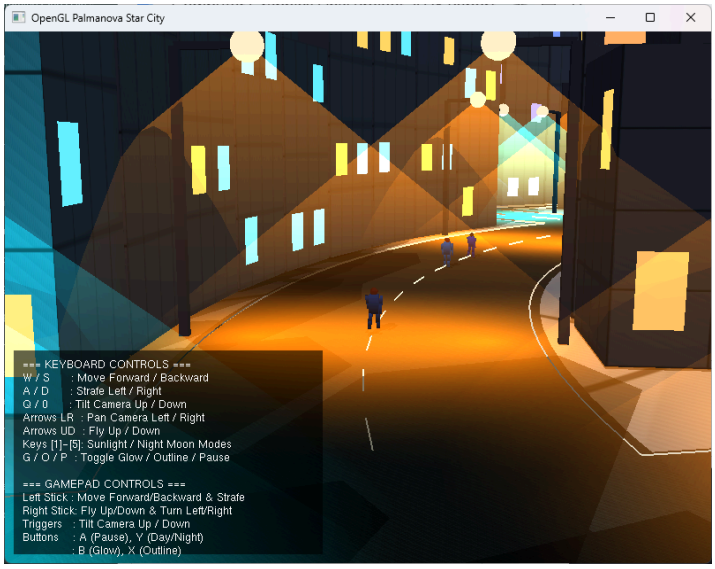
- **Animation 2:** 8 AI delivery drones follow dynamic 3D orbital paths using sine and cosine waves.



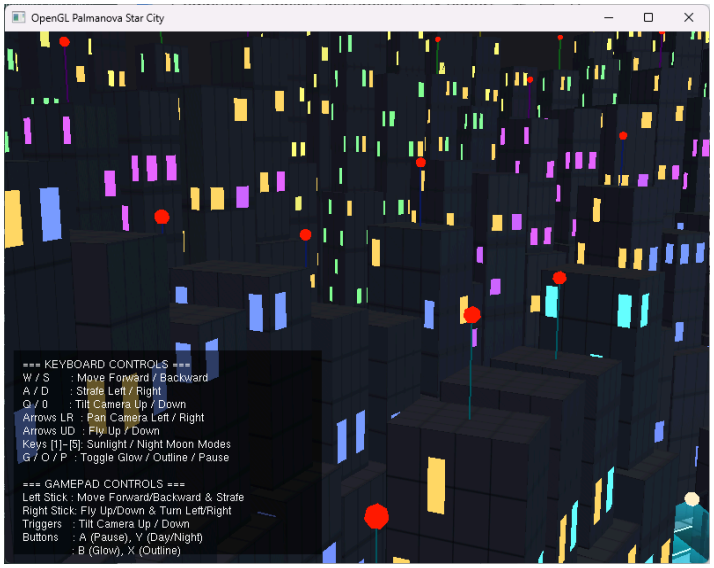
- **4. Light and Shadow (10%)**:
  - The project implements 7 distinct hardware light sources, including global directional light, a neon point light, and dynamic street spotlights.
  - Visible planar shadows are rendered using matrix transformations.



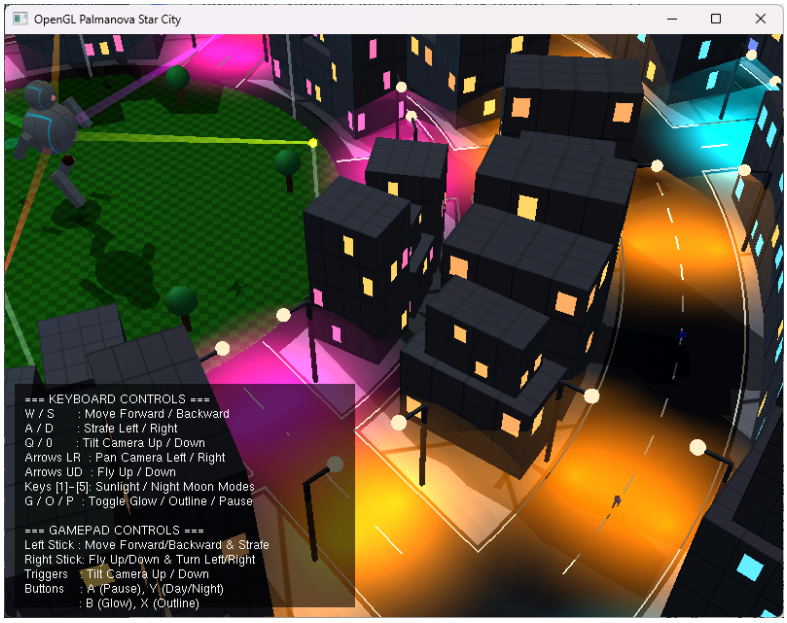
Moon Light: Directional light



Street Lamp: Spotlight



Neon: Point light



Shadows

v. Technical Challenges

This section outlines the difficulties encountered during development and the solutions adopted.

- **Challenge 1: Z-Fighting (Flickering Pavements)**
  - *Difficulty:* The ground terrain, roads, and avenues were all placed at the exact same height (Y = -0.4), causing the GPU depth buffer to struggle and create severe rendering flickering.
  - *Solution:* Implemented microscopic height offsets to layer the objects cleanly. Base

concrete was set to Y = -0.4, road strips to Y = -0.398, road markings to Y = -0.395, and skyscraper bases to Y = -0.39 to eliminate visual glitches.

- **Challenge 2: Multi-Overlay Shadow Darkness (Double Blending)**
  - *Difficulty:* Overlapping 3D objects projecting planar shadows onto the same ground plane resulted in pitch-black, multiply-blended overlap regions.
  - *Solution:* Integrated the Stencil Buffer. Stencil testing writes a 1 for drawn shadow pixels and rejects subsequent fragments if the stencil value is already non-zero, resulting in uniform soft shadows.
- **Challenge 3: Hardware Light Source Limit**
  - *Difficulty:* Standard OpenGL limits applications to only 8 hardware lights, which was not enough for a massive city requiring 180 streetlights.
  - *Solution:* Developed a Dynamic Spotlight Allocation System. At every frame, the program calculates the distance from the player camera to all 180 lamps, sorts them, and dynamically binds the 4 closest lamps to hardware channels.
- **Challenge 4: Backface Culling & Winding Order**
  - *Difficulty:* The circular central park grass disc became invisible from certain viewing angles because the vertices were drawn in a clockwise order.
  - *Solution:* Rewrote the rendering loop with a negative angle step to produce a counter-clockwise (CCW) winding order, properly satisfying the backface culling requirements.